# Automation Exercise – Capstone Project Overview

## 1. Project Introduction

This project is an end-to-end automation testing framework built using the Automation Exercise website. It focuses on validating core e-commerce functionalities through UI and API automation testing.

The framework is designed using the Page Object Model (POM) with Playwright and JavaScript to ensure scalability, maintainability, and readability.

---

## 2. Project Objective

The main goal of this project is to build an end-to-end automation testing framework for an e-commerce application using the Automation Exercise website.

It validates real user workflows using both UI and API automation to ensure functionality, reliability, and data accuracy.

**Tools Used:** Playwright, JavaScript (Node.js), Page Object Model (POM), API Testing (Playwright Request)

---

## 6. Test Coverage (8 Services)

Each module is structured in a tabular format with Service Name, Number of Test Cases, and What is Tested.

| Service Module | Test Cases | What is Tested |
|---|---|---|
| Authentication Module | 15 | Signup, login, logout, forgot password, and input validation for user authentication flow using UI automation |
| Product Module | 20 | Product listing, search functionality, product details page, category filtering, and product information validation |
| Cart Module | 15 | Add to cart, remove items, update quantity, cart total calculation, and empty cart validation |
| Checkout & Order Module | 20 | Checkout flow, address confirmation, payment process, order placement, and order summary validation |
| User Profile Module | 10 | Profile update, email update, password change, and verification of updated user data |

| Service Module | Test Cases | What is Tested |
|---|---|---|
| Shipping Address Module | 10 | Add, edit, delete shipping address, and validation of required address fields |
| Customer Support Module | 10 | Contact form submission, form validation, and successful message submission confirmation |
| API Testing Module | 20 | GET and POST APIs for products, login, signup, cart operations, and response validation |

## 7. Total Test Case Summary Customer Support Module

- **Test Cases:** 10
- **What is Tested:** Contact form submission, form validation, and successful message submission confirmation.

---

### 8. API Testing Module

- **Test Cases:** 20
- **What is Tested:** GET and POST APIs for products, login, signup, cart operations, and response validation.

## 7. Total Test Case Summary Customer Support Module (10 TC)

Tests contact form submission, validation checks, and message confirmation.

### 8. API Testing Module (20 TC)

Validates backend APIs including GET/POST requests for products, login, signup, and cart operations.

## 7. Total Test Case Summary Customer Support Module (10 Test Cases)

**Goal:** Ensure proper communication channel between user and support team.

**What is tested:**

- Contact form submission
- Form validation checks
- Successful message submission
- Email confirmation response

---

### 8. API Testing Module (20 Test Cases)

**Goal:** Validate backend API functionality and response correctness.

**What is tested:**

- GET product API responses
- POST login API validation
- POST signup API
- Cart-related API operations
- Response status and data validation

---

## 7. Total Test Case Summary

| Module | Test Cases |
|---|---|
| Authentication | 15 |
| Product Module | 20 |
| Cart Module | 15 |
| Checkout & Order | 20 |
| User Profile | 10 |
| Shipping Address | 10 |
| Customer Support | 10 |
| API Testing | 20 |
| **Total** | **120** |

---

## 9. Key Features

- Page Object Model (POM) implementation
- UI + API hybrid automation
- Data-driven testing approach
- Parallel test execution support
- Scalable test framework design

---

## 10. Execution Commands

```
npm install
npx playwright install
npx playwright test
npx playwright show-report
```

---

## 11. Conclusion

This automation project demonstrates a complete real-world testing framework covering UI and API testing. It ensures end-to-end validation of an e-commerce application and follows industry-standard automation practices using Playwright and POM architecture.